Proxmox VE Introduction

Day 3: Storage, Backup & Operations

Day 3 — Agenda

1. Recap: Day 2 Key Points
2. Storage Backends Overview — Directory, LVM, ZFS, Ceph
3. Backup & Restore with Proxmox Backup Server
4. Scheduled Backups & Retention Policies
5. Basic High Availability Concepts
6. Updates, Upgrades & the Subscription Model
7. Monitoring — Metrics & Notifications
8. Troubleshooting Common Issues
9. Best Practices for Small/Medium Deployments
10. Course Wrap-up & Next Steps

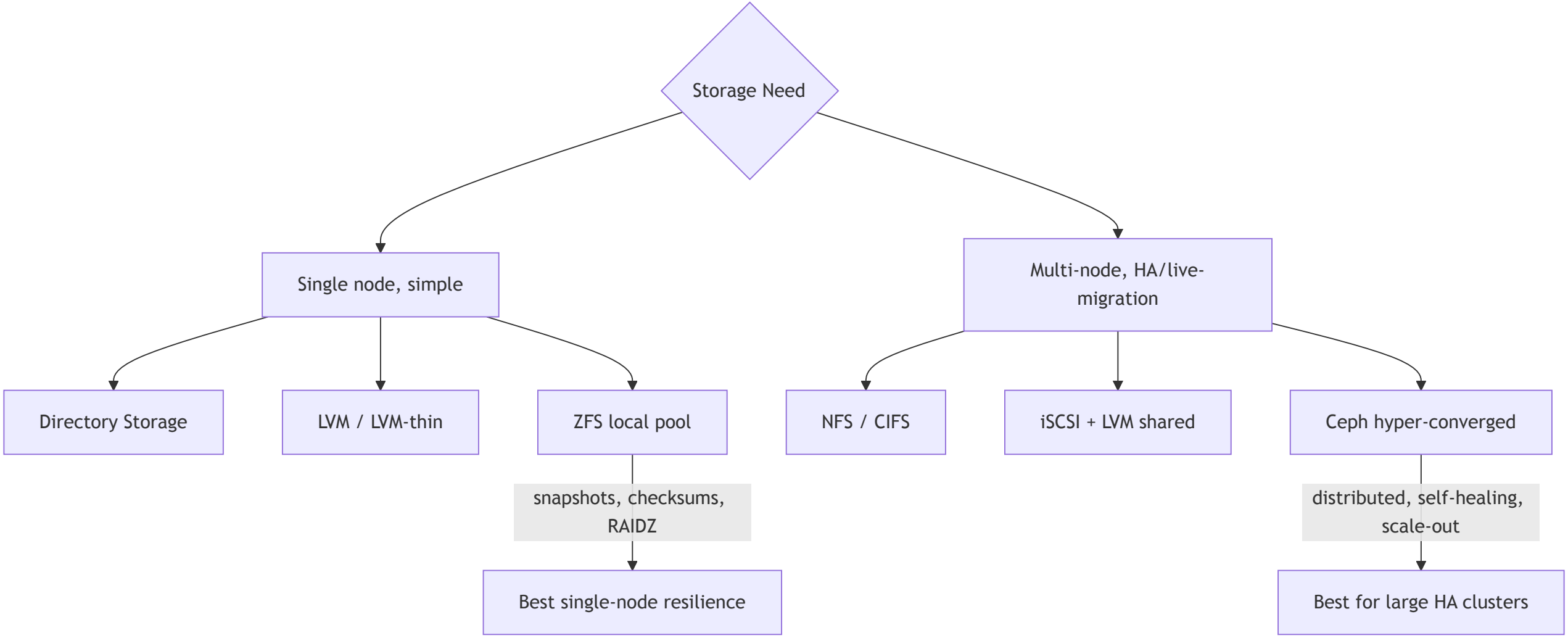
1. Recap: Day 2 Key Points

- VMs run as **QEMU processes** accelerated by **KVM**; containers share the host kernel via LXC
- Templates + linked clones = fast, disk-efficient provisioning
- Live migration and snapshots rely on the same block/memory primitives
- **VirtIO** drivers + **qemu-guest-agent** are essential for performance and backup consistency

Today: how to **store, protect, and operate** everything we built on Day 2.

2. Storage Backends Overview

Choosing a storage backend is one of the most consequential early decisions in any Proxmox VE deployment — it determines whether you get live migration, snapshots, and HA "for free."

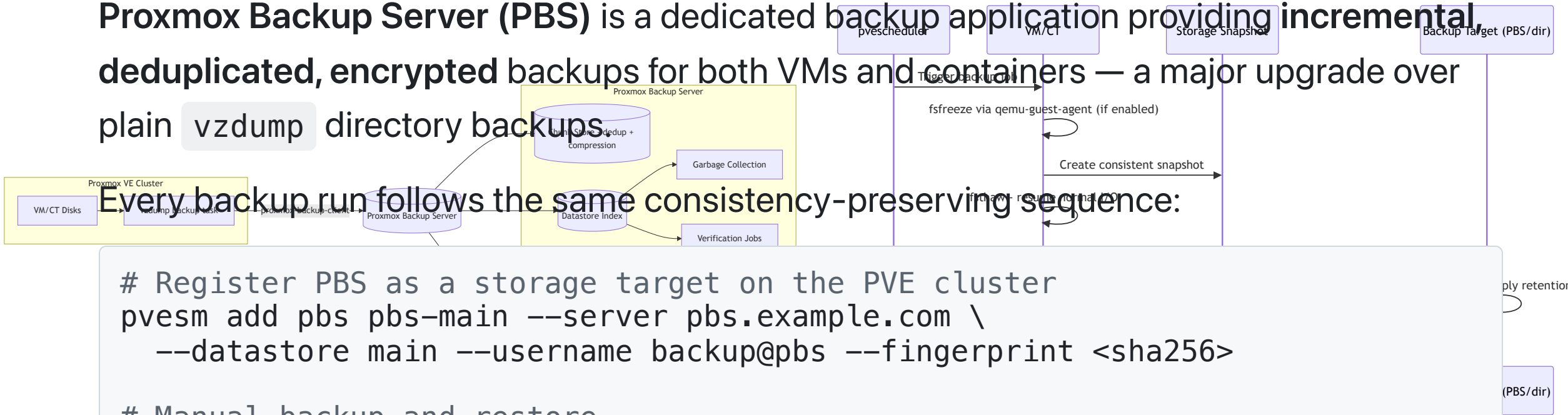

Storage Need
Single node, simple
Multi-node, HA/live-migration
Directory Storage
LVM / LVM-thin
ZFS local pool
NFS / CIFS
iSCSI + LVM shared
Ceph hyper-converged
snapshots, checksums, RAIDZ
distributed, self-healing, scale-out
Best single-node resilience
Best for large HA clusters

```
pvesm status
pvesm add zfspool tank --pool rpool/data --content images,rootdir
pvesm add nfs backup-nfs --server 10.10.10.20 --export /export/backup --content backup
```

Rule of thumb: single node with no HA needs → local ZFS. Multi-node cluster wanting HA/live-migration without dedicated storage hardware → **Ceph**. Existing SAN/NAS → NFS or iSCSI+LVM.

3. Backup & Restore with Proxmox Backup Server

Proxmox Backup Server (PBS) is a dedicated backup application providing **incremental, deduplicated, encrypted** backups for both VMs and containers — a major upgrade over plain `vzdump` directory backups.



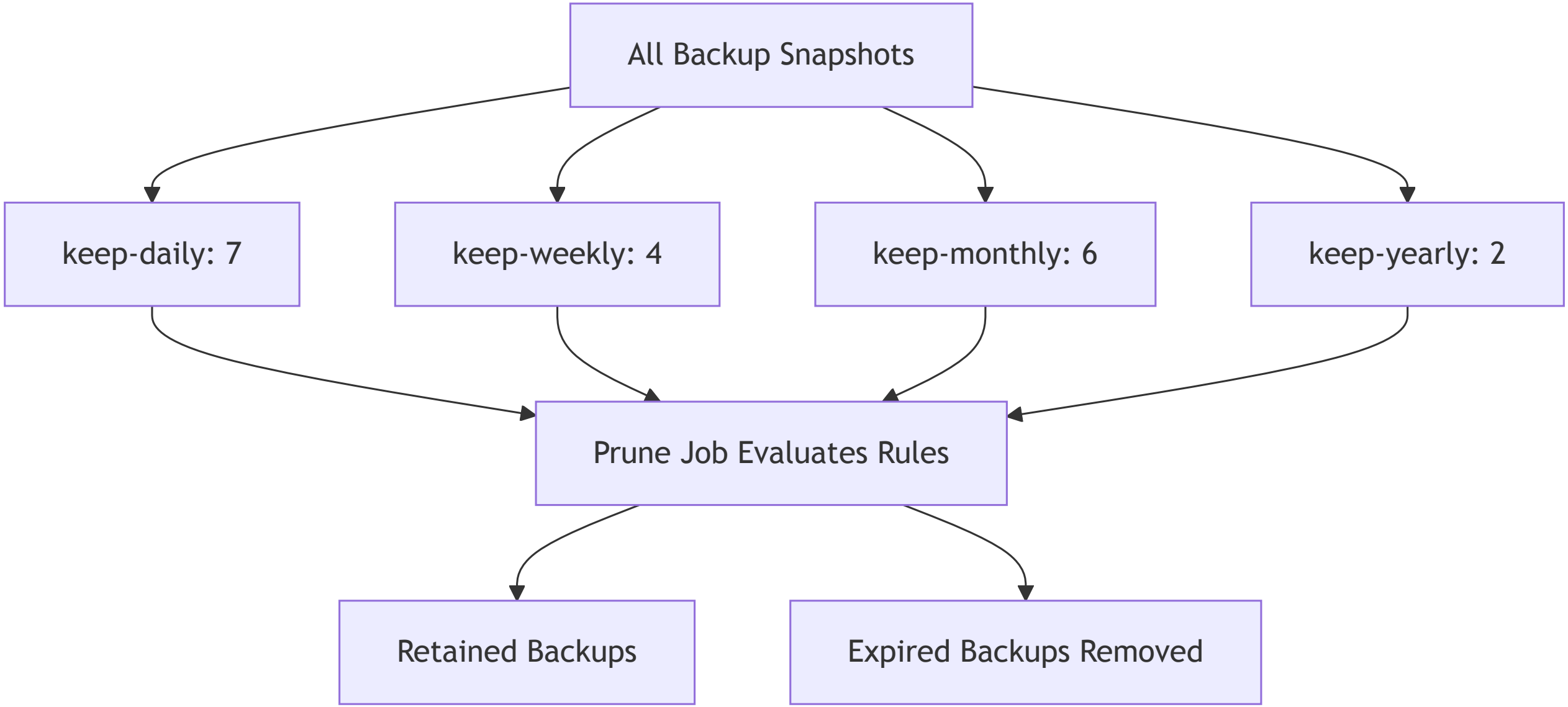
Every backup run follows the same consistency-preserving sequence:

```
# Register PBS as a storage target on the PVE cluster
pvesm add pbs pbs-main --server pbs.example.com \
  --datastore main --username backup@pbs --fingerprint <sha256>

# Manual backup and restore
vzdump 100 --storage pbs-main --mode snapshot
qmrestore pbs-main:backup/vm/100/2026-08-01T02:00:00Z 100 --storage local-zfs
```

4. Scheduled Backups & Retention Policies

Backup jobs are defined once (UI: *Datacenter* → *Backup*, or `/etc/pve/vzdump.cron` - managed) and run unattended on a schedule. **Retention** rules then prune old backups automatically:


All Backup Snapshots
keep-daily: 7
keep-weekly: 4
keep-monthly: 6
keep-yearly: 2
Prune Job Evaluates Rules
Retained Backups
Expired Backups Removed

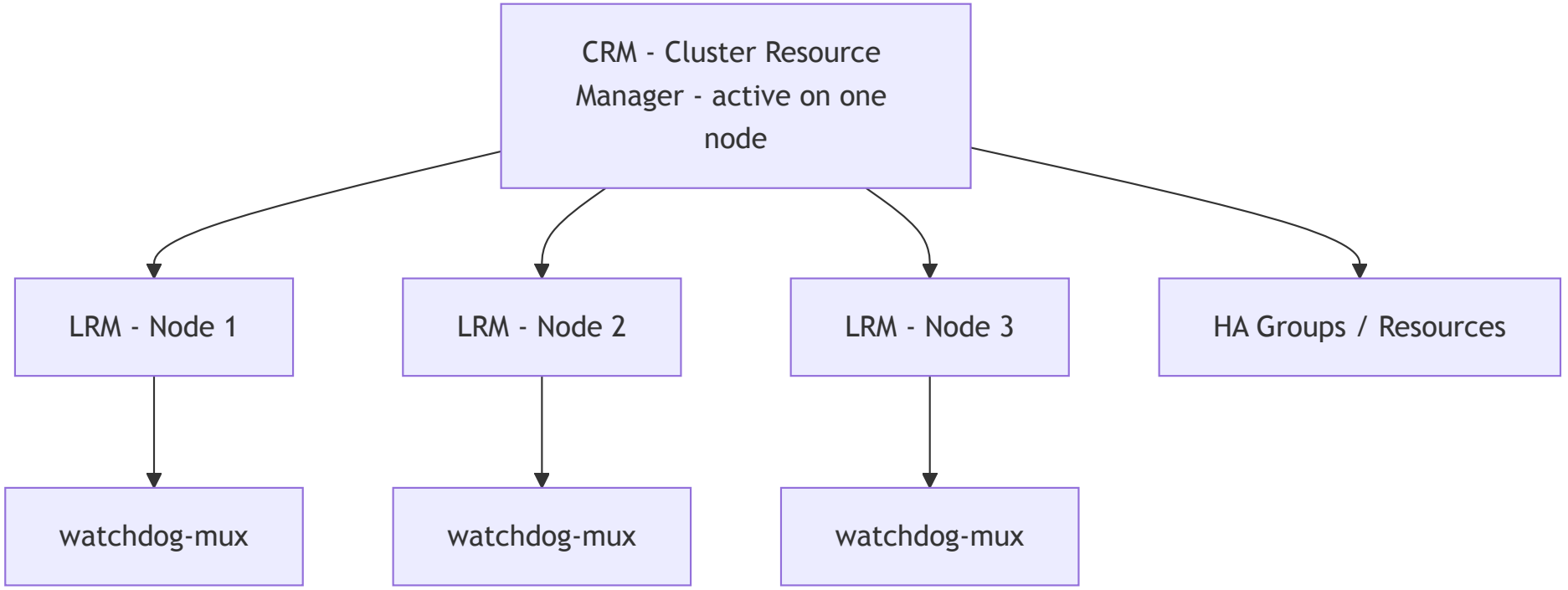
```
# Retention example: keep granular recent history, sparse long-term history
proxmox-backup-client prune --group vm/100 \
  --keep-daily 7 --keep-weekly 4 --keep-monthly 6 --keep-yearly 2

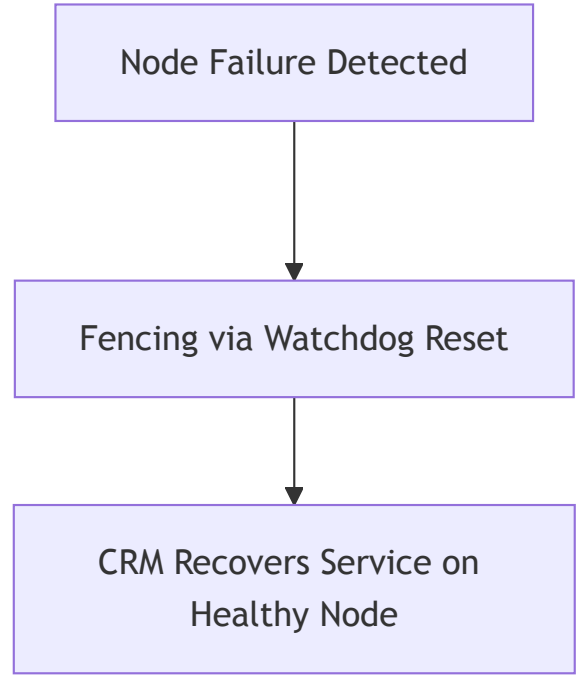
# PBS verifies stored chunks for silent bit-rot detection
proxmox-backup-manager verify-job create daily-verify --schedule daily
```

Always pair a retention policy with **PBS remote sync** (`sync-job`) to a second datastore — ideally offsite — for genuine disaster-recovery coverage.

5. Basic High Availability Concepts

Proxmox VE HA automatically restarts a VM/CT on a healthy node if its current node fails, using a **watchdog-based fencing** mechanism to guarantee the failed node is truly down before recovery starts.


CRM - Cluster Resource Manager - active on one node
LRM - Node 1
LRM - Node 2
LRM - Node 3
HA Groups / Resources
watchdog-mux
watchdog-mux
watchdog-mux


Node Failure Detected
Fencing via Watchdog Reset
CRM Recovers Service on Healthy Node

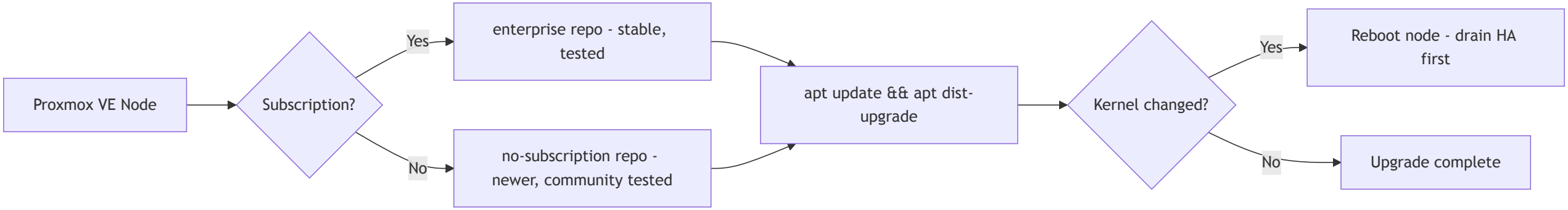
```
# Add a VM to HA management with a group and restart policy
ha-manager add vm:100 --group prod-nodes --max_restart 3 --max_relocate 2
ha-manager status
```

HA **requires shared storage** (Ceph, NFS, iSCSI+LVM) — a VM whose disk only exists on local storage cannot be restarted on a different node. Deep-dive HA design (fencing edge cases, affinity rules, DR) is covered in the Architect course.

6. Updates, Upgrades & the Subscription Model

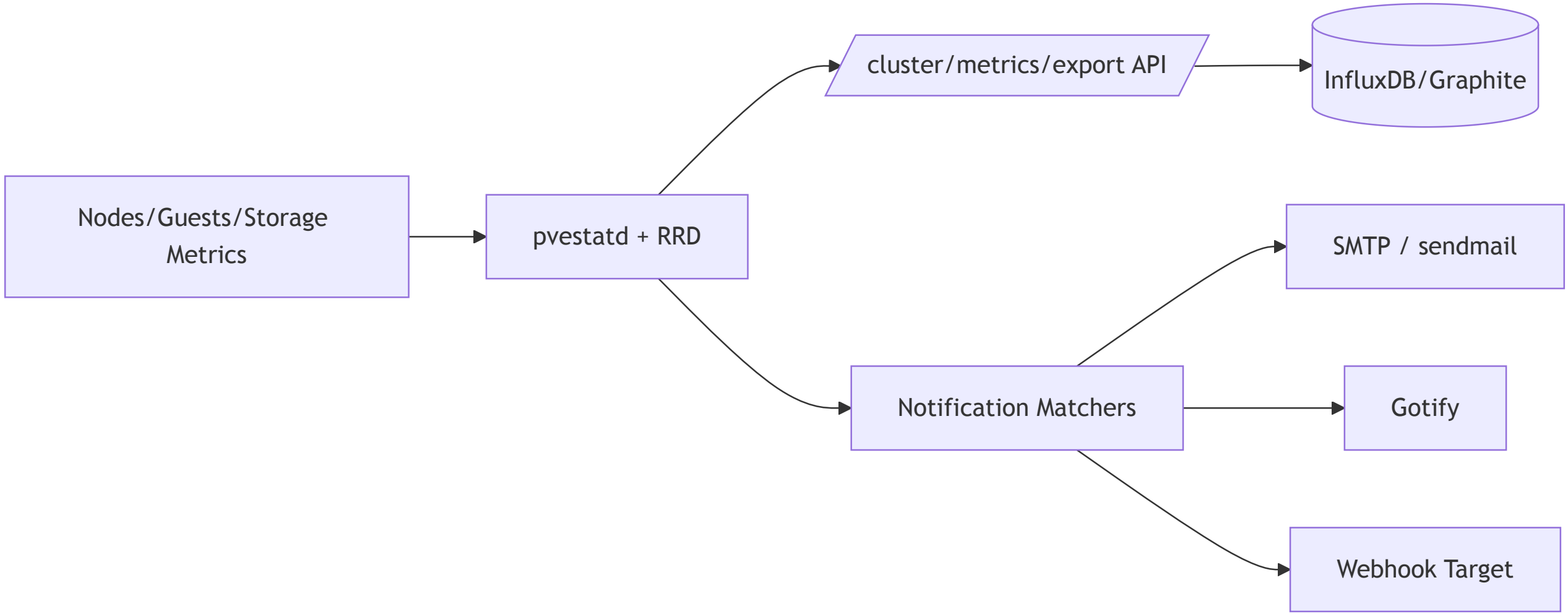
Repository	Stability	Cost
`enterprise`	Most tested, delayed	Requires paid subscription
`no-subscription`	Current, community-tested	Free
`pvetest`	Bleeding edge	Free, **not for production**

```
apt update && apt full-upgrade -y
pveversion -v
# Kernel change? Reboot with HA-aware node maintenance:
ha-manager crm-command node-maintenance enable pve1
reboot
```


Proxmox VE Node
Subscription?
Yes
No
enterprise repo - stable, tested
no-subscription repo - newer, community tested
apt update && apt dist-upgrade
Kernel changed?
Yes
No
Reboot node - drain HA first
Upgrade complete

7. Monitoring — Metrics & Notifications

Proxmox VE collects metrics locally via `pvestatd` /RRD for the built-in UI graphs, and can also export them to external time-series systems. Separately, a flexible **notification system** routes events (backup failures, replication errors, HA state changes) to real alerting channels.


Nodes/Guests/Storage Metrics
pvestatd + RRD
cluster/metrics/export API
InfluxDB/Graphite
Notification Matchers
SMTP / sendmail
Gotify
Webhook Target

```
pvesh set /cluster/metrics/server/influxdb --type influxdb \
  --server influx.example.com --port 8089

# Notifications: matchers route by event type to a target (email, Gotify, webhook)
pvesh create /cluster/notifications/endpoints/gotify \
  --name ops-gotify --server https://gotify.example.com --token <token>
```

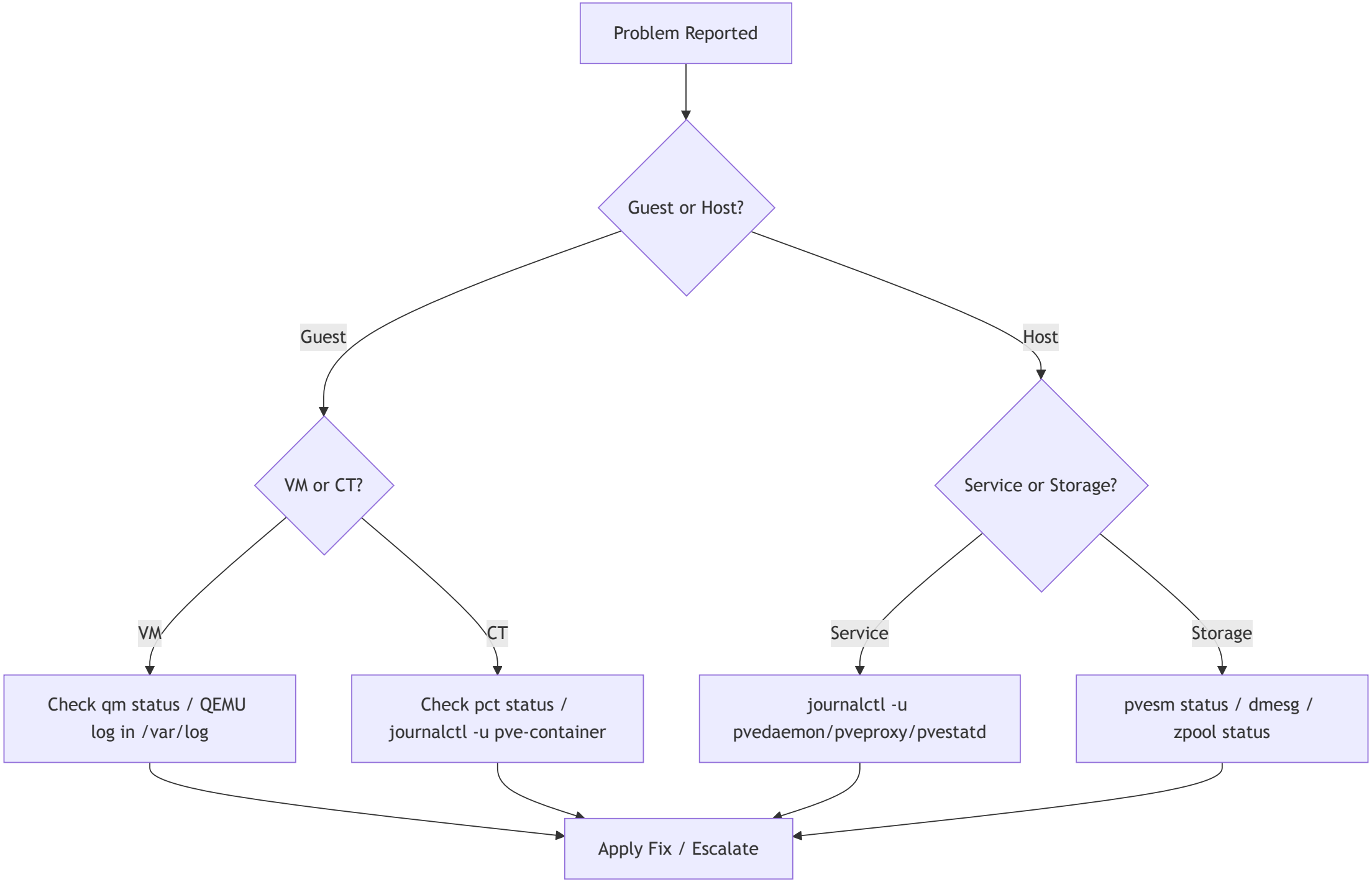
8. Troubleshooting Common Issues

A consistent first-triage question: **is the problem in the guest, or on the host?**

```
# Guest issues
qm status 100 && qm showcmd 100
journalctl -u pve-container@300 -f

# Host/service issues
journalctl -u pvedaemon -u pveproxy -u pvestatd --since "1 hour ago"

# Storage issues
pvesm status
zpool status -v
dmesg -T | tail -50
```


Problem Reported
Guest or Host?
Guest
Host
VM or CT?
Service or Storage?
VM
CT
Service
Storage
Check qm status / QEMU log in /var/log
Check pct status / journalctl -u pve-container
journalctl -u pvedaemon/pveproxy/pvestatd
pvesm status / dmesg / zpool status
Apply Fix / Escalate

9. Best Practices for Small/Medium Deployments

- Use **ZFS with a mirror or RAIDZ** for local storage resilience, even on a single node
- Always configure **at least one off-node backup target** (PBS is strongly preferred)
- Keep nodes on the **no-subscription** repo current unless you hold a paid subscription
- Enable the **firewall** at the datacenter level even for "internal" clusters
- Use **API tokens**, not shared root passwords, for any automation (Terraform/Ansible/CI)
- Document your network plan (bridges/VLANs) before scaling past 1 node

10. Course Wrap-up & Next Steps

You now have a working foundation across virtualization concepts, VM/container operations, storage, backup, and day-2 operations.

Where to go from here:

- **Proxmox VE Architect** — cluster design, storage architecture (ZFS/Ceph deep-dive), advanced networking/SDN, HA/DR at scale
- **Proxmox VE Security** — hardening, RBAC deep-dive, network segmentation, compliance (ISO 27001, CIS Benchmarks)

Thank you for attending — questions welcome!